TOM - Muon Tomography

Signature Sheet

Student's Name _____ Partner's Name _____

Suggested reading to start with (see the end of the manual for more):

1. 111B TOM manual
2. 111B MUO manual (if you did not do MUO, you should read through this manual for more context and consider talking through the additional prelab questions with a GSI)
3. Alvarez et al., Search for Hidden Chambers in the Pyramids
4. *Applications of Cosmic Ray Muons*

Pre-Lab Discussion Questions

It is your responsibility to discuss this lab with an instructor before your first day of your scheduled lab period. This signed sheet must be included as the first page of your report. Without it you will lose grade points. You should be prepared to discuss at least the following before you come to lab:

1. If you are interested in going to a building other than Evans Hall and/or Cory Hall, at least two days before the start of the lab, check in with instructors about whether access needs to be granted to you for any of the spaces you will be taking the detector cart to.
2. What is a muon? How and where are the muons in this experiment produced? Introduce different types of muon decay. **If you have done MUO, you may skip this question.**
3. How do we capture the muon and convert the decay process into an electrical signal so that we can do statistics? **If you have done MUO, you may skip this question.**
4. What do we call a 'coincidence' and a 'stoppage' in this lab? What is the difference, and when does it matter?
5. What is muon tomography or muography? How can it be used to 'map' or 'image' a building or object? What sorts of algorithms are used in tomographic imaging, and in what other contexts is tomography used?
6. One important aspect of this experiment is setting the threshold for considering a pair of pulses a coincidence. Given two pulses with heights y_1 and y_2, and peak times t_1 and t_2, what are possible heights for the product of the two pulses? Assume the pulses are Gaussian. (Hint: you should be showing a plot; consider Δy and Δt.) What does this tell you about setting the trigger level for coincidences (i.e., given this plot, what additional information do you need about how pulses look on the two detectors in order to set the coincidence threshold)?
7. Given the geometry of the two detectors "Bra" (1'×1'×1') and "Ket" (4"×1'×1'), calculate the number of cosmic rays per minute which enter each detector.
8. At what rate will random coincidences occur given some trigger level and the count rates you calculated in the previous question? (Hint: you may want to reference EAX.)

9. How does count rate change per story of a building you are in? (It may be helpful to note that cosmic muons lose energy via ionization at approximately $\frac{dE}{dx} \approx 2$ MeV/(g/cm^2), and that a typical concrete floor slab has density $\rho \approx 2.3$ g/cm^3 and is 15–20 cm thick. You may see Figure 5 of Rossi for the cosmic muon differential spectrum and use an estimate at low energies.) Then, given two measurements on two different floors, how long of a measurement do you need for the difference in count rate to be statistically significant (it can be convenient to consider percent difference between floors)? For this, you can use a z-statistic or another statistical test. Assume you are only looking at coincidences (not stoppages), and you may assume a coincidence rate $R_C \approx 1$Hz, but you should check this value once you start the experiment. How about for a three floor separation?

10. Note that the number of muons that stop in the detector is dependent only on the mass of the detector (see Figure 6 of Rossi, linked in the references in the TOM manual). Assume the density of the mineral oil is 0.8 g/cm^3 and that we are looking for stoppages in 'Bra' (the larger detector). If you are now looking for stoppages instead of just coincidences, how does this change your answer to the previous question? (You may want to consider a 5 floor separation as well.)

11. What must you do whenever you take the cart out of the lab?

Staff Signature _____ Date _____

Completed before the first day of lab? (Circle one) Yes / No

Mid-Lab Discussion Questions

1. By day 4, you should have a plan for how many locations and angles at each location you will need to take measurements at in order to perform a tomographic reconstruction of Evans or Cory Hall, and some rough calculations of what angular resolution you'll be able to achieve with your detector. Part of this plan should be your timeline for how long your measurements will take and when you'll take each! Discuss your plan and the answers to the questions for planning the angular dependence measurements with a GSI/instructor.

Staff Signature _____ Date _____

Completed by day 4 of lab? (Circle one) Yes / No

Checkpoint Signatures

1. <u>Pulse thresholds</u>

   Staff Signature _____

2. <u>Coincidence detection threshold</u>

Staff Signature _____

3. Material absorption coefficients

Staff Signature _____

4. Coincidences vs. stoppages

Staff Signature _____

5. Vertical count rate

Staff Signature _____